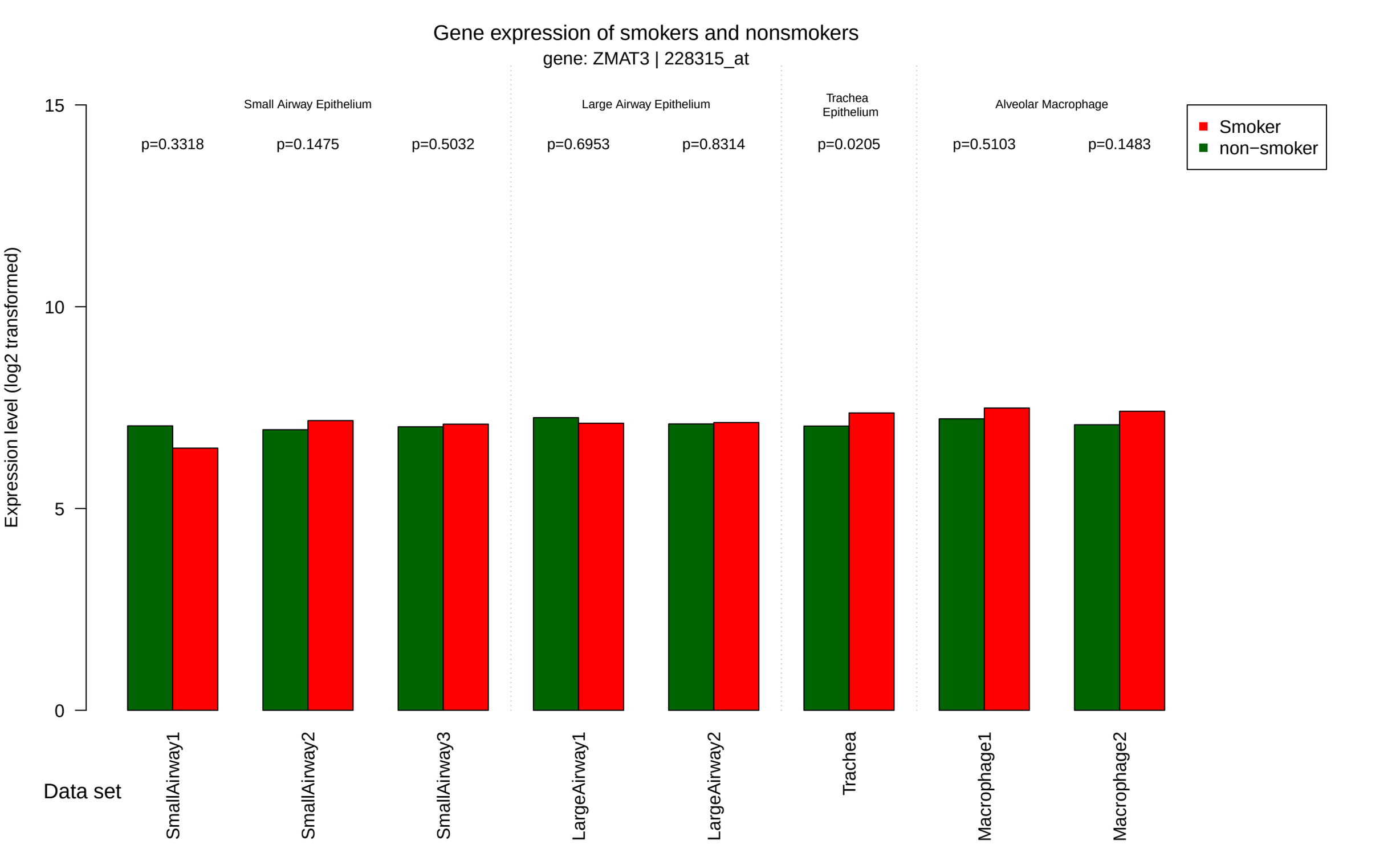
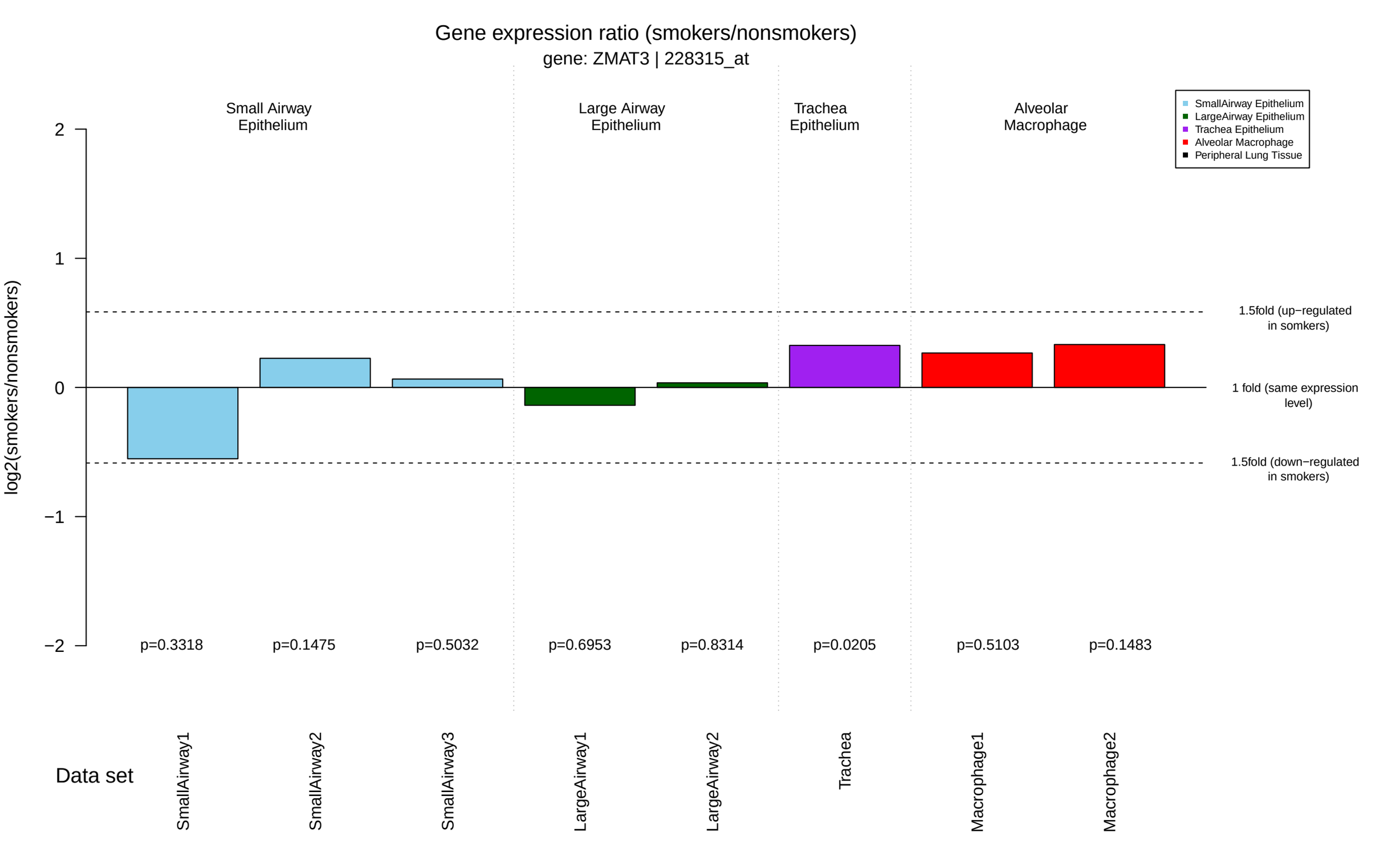
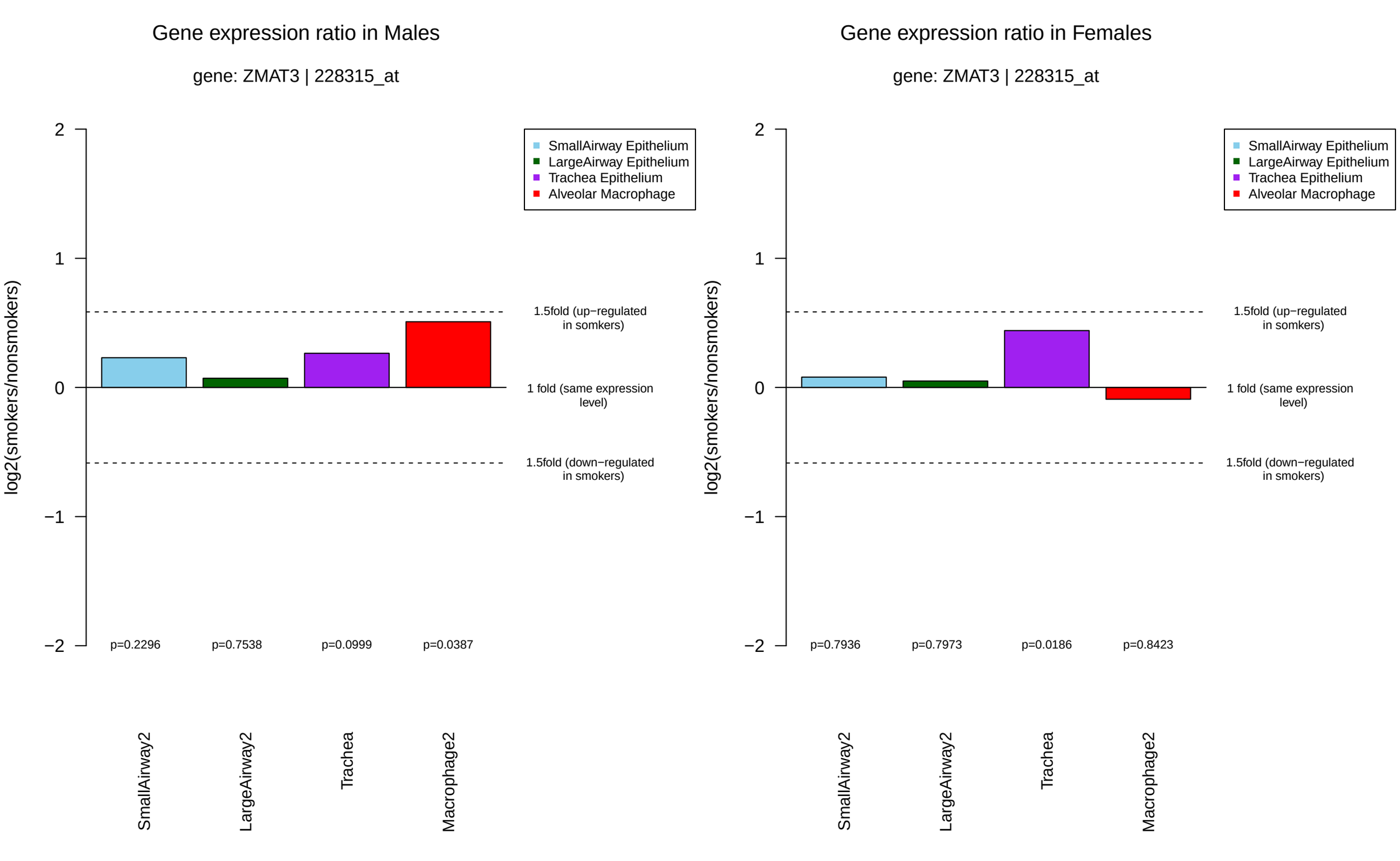
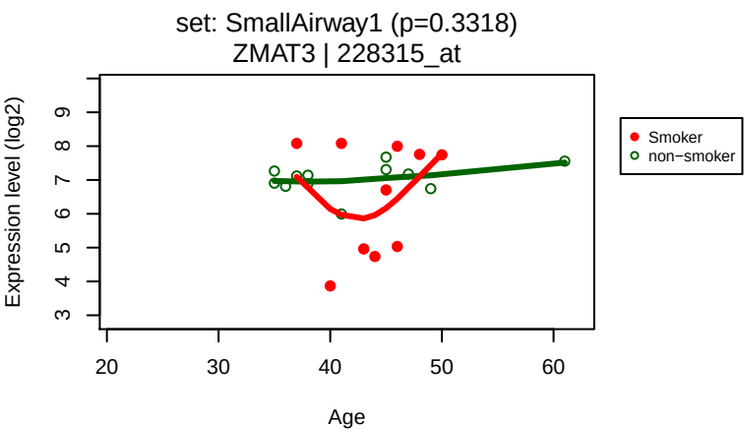
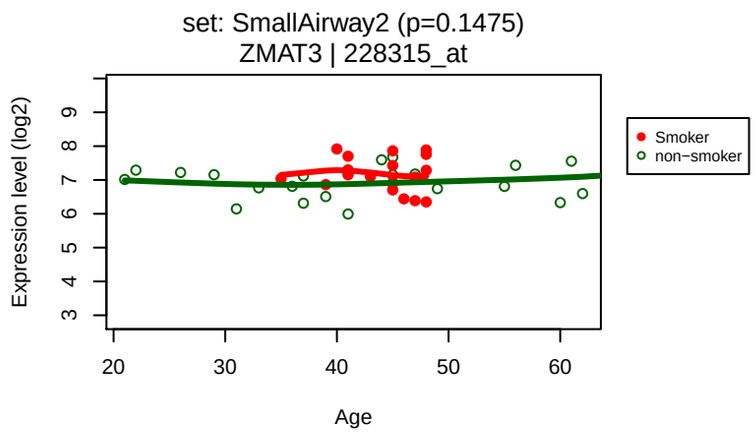
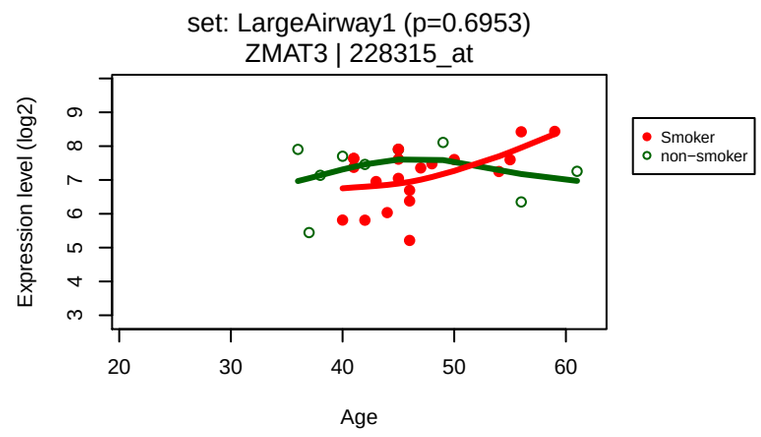

Gene expression of smokers and nonsmokers
gene: ZMAT3 | 228315_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3318
p=0.1475
p=0.5032
p=0.6953
p=0.8314
p=0.0205
p=0.5103
p=0.1483
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

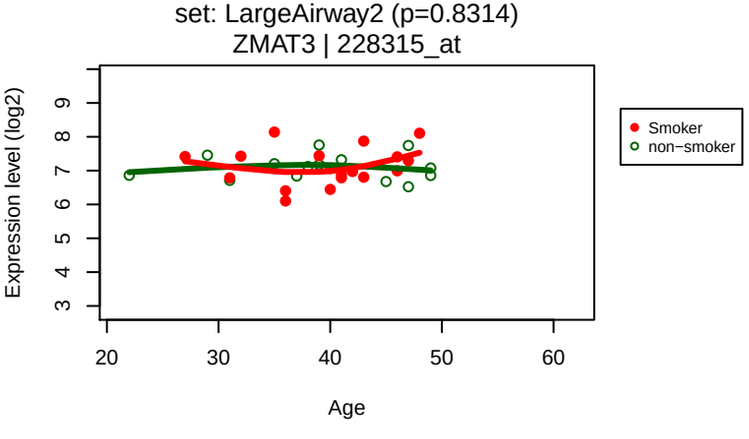

Gene expression ratio (smokers/nonsmokers)
gene: ZMAT3 | 228315_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3318
p=0.1475
p=0.5032
p=0.6953
p=0.8314
p=0.0205
p=0.5103
p=0.1483
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

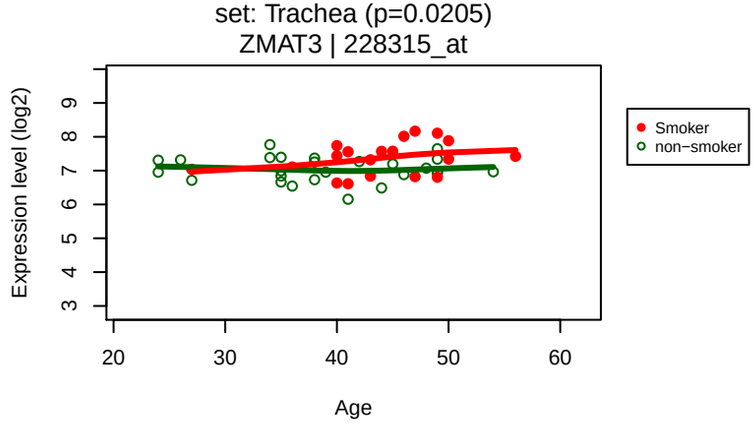

Gene expression ratio in Males
gene: ZMAT3 | 228315_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2296
p=0.7538
p=0.0999
p=0.0387
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ZMAT3 | 228315_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7936
p=0.7973
p=0.0186
p=0.8423
SmallAirway2
LargeAirway2
Trachea
Macrophage2

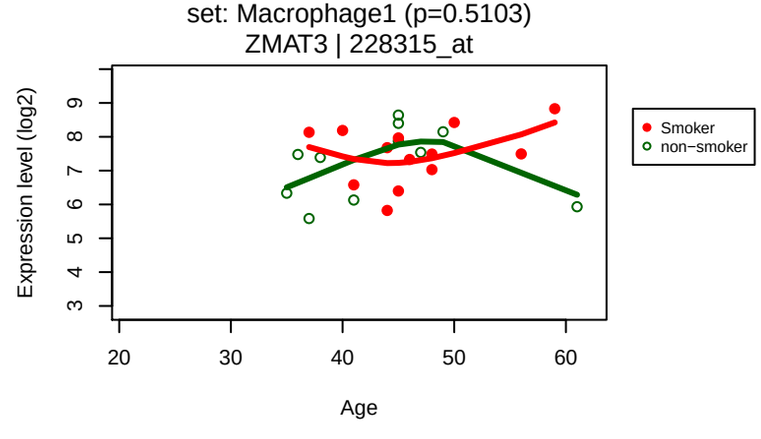

set: SmallAirway1 (p=0.3318)
ZMAT3 | 228315_at
Expression level (log2)
Age
Smoker
non−smoker

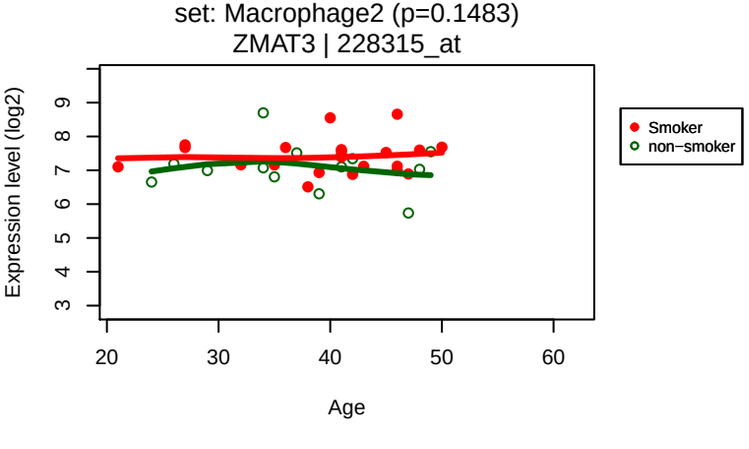

set: SmallAirway2 (p=0.1475)
ZMAT3 | 228315_at
Expression level (log2)
Age
Smoker
non−smoker

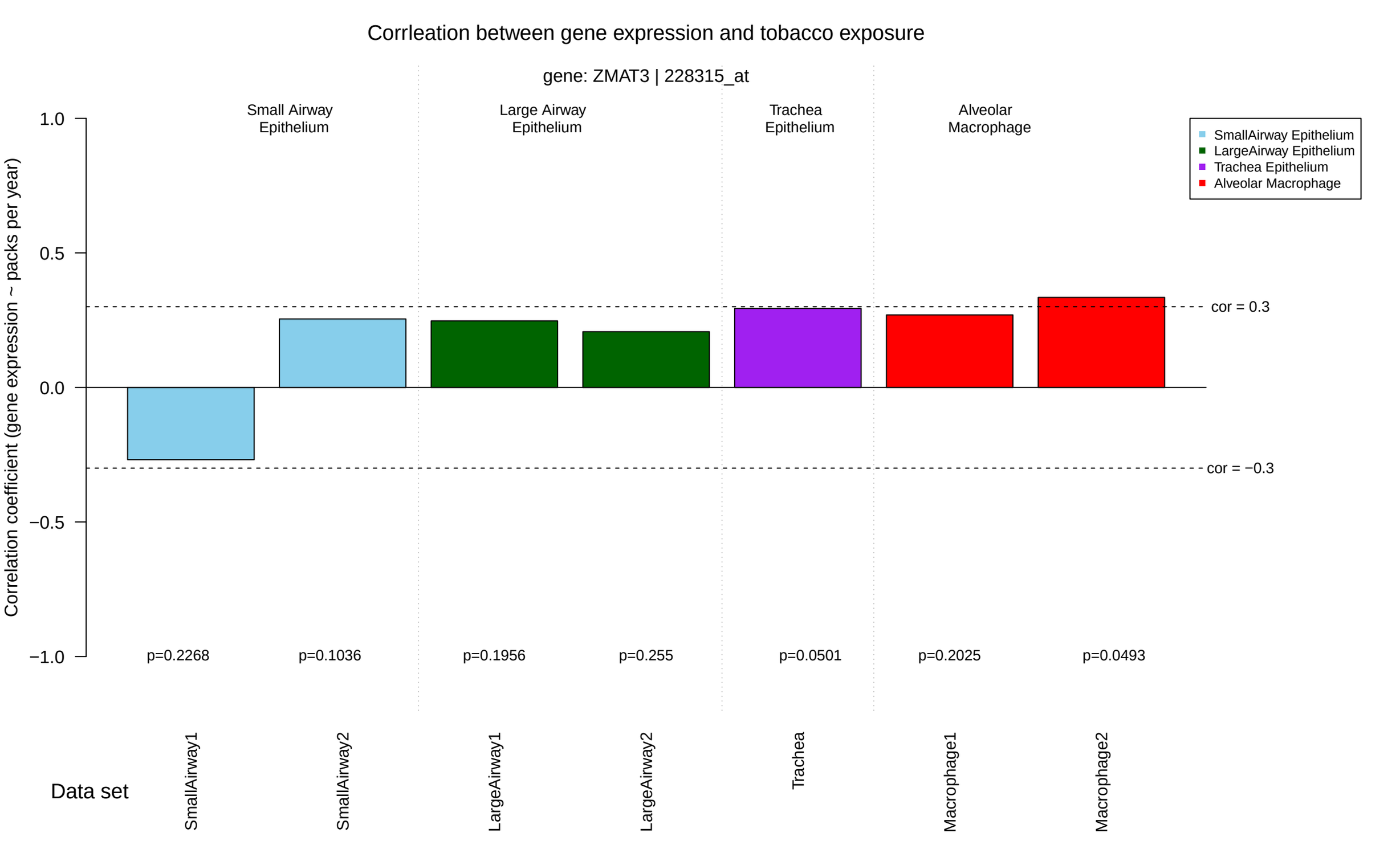

set: LargeAirway1 (p=0.6953)
ZMAT3 | 228315_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8314)
ZMAT3 | 228315_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0205)
ZMAT3 | 228315_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5103)
ZMAT3 | 228315_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1483)
ZMAT3 | 228315_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ZMAT3 | 228315_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2268
p=0.1036
p=0.1956
p=0.255
p=0.0501
p=0.2025
p=0.0493
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2